

MSc Mathematical and Computational Finance

Statistics and Financial Data Analysis - Problem Sheet 2

1. Data Analysis Report: restrict answers for this session to no more than 2 pages, clearly stating the problem you are examining, describing the data, labelling your graphs and backing up your conclusions with evidence. Your research may be extensive, however, the report should focus on describing the models considered, summarising your key findings with relevant data. The report should not be reduced to a 'cut-and-paste' exercise of the outputs of all your code.
 - (a) The following csv files `IR.csv`, `Repossession.csv`, `Unemployment.csv` were created using data available at:

`https://www.gov.uk/government/statistical-data-sets/
live-tables-on-repossession-activity`
`https://www.bankofengland.co.uk/boeapps/database/Bank-Rate.asp`
`http://www.ons.gov.uk/employmentandlabourmarket/peoplenotinwork`
 - (b) Investigate the historical relationship in the UK between the bank base interest rates, the rate of unemployment, and the rate of mortgage repossession. Write a brief report of your findings, including the main models considered and their assumptions and how they were verified.
 - (c) Create a single dataframe `df` in Python, aligning the dates for the three datasets. Display the first three and last three elements of your dataframe using the commands `df.head(3)` and `df.tail(3)`. Display this output in your report (insert the picture of the output generated in Python).
 - (d) Note: you should try simple regression models, polynomial regression and piecewise polynomial regression/splines (no need for kernel regression). Consider which features to include and if normalising the data improves the model. Also use appropriate model selection criteria to support your conclusions.
2. Download the files `Python_DataQ2` and save it as a dataframe `df_WLS`.
 - (a) Regress `y_data` on `x` using a standard OLS linear regression. Display the fitted values graph against the original data as well as the residual against the inputs `x`. What are your observations regarding the residuals?
 - (b) Use the weights from the DataFrame `df_WLS['weights']` to create a Weighted Linear Regression. These weights are proportional to the standard deviations of distribution of the errors. Display the estimated coefficients from WLS alongside the ones from OLS and their respective standard deviations.
 - (c) Plot the prediction interval, alongside the fitted values and original data in a single graph. What can you observe?

3. We seek to fit a model of the form $Y_i = f(x_i) + \sigma\epsilon_i$. Write a Python script/notebook which searches for the optimal spline approximation to this model (with knots given by the quantiles of x), using the AIC as an optimality criterion. Your script should take vectors y and x as inputs, and return the number of knots in the optimal model. You may assume that the optimal number of knots is less than `sqrt(length(x))`, and that the vector x has at least this many distinct quantiles.